Support Vector Machine Implementation

1. You must build a full SVM-based classification workflow that includes:

- Dataset selection and loading
- Data preprocessing
- Model training
- Hyperparameter tuning (kernel, C, gamma, degree)
- Evaluation and visualization

2. Perform Essential Data Preprocessing

- Handle missing values
- Encode categorical features
- Normalize/standardize numerical features
- Split into train/test/validation sets

3. Dataset

- Download any dataset from Kaggle or other open source
- Upload the dataset to Google Drive

4. Model Implementation in Google Colab

- Data loading
- Preprocessing
- SVM model training
- Hyper-parameter tuning for optimization
- Generating evaluation metrics

5. Include the Following Screenshots

- Confusion matrix (heatmap preferred)
- Evaluation metrics: Accuracy, Precision, Recall, F1-score, AUC
- ROC Curve
- Sample predictions (actual vs predicted + probability scores)
- 2D Classification plot

6. Submission

- Colab link
- Dataset link

- Required screenshots